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Wafer assembly for electrical connector assemblies

Abstract

A wafer assembly includes a leadframe having signal contacts and cables with signal conductors terminated to the corresponding signal contacts. Each cable includes a cable shield providing shielding for the signal conductors. The wafer assembly includes a wafer body holding the signal contacts. The wafer assembly includes first and second ground frames coupled to the first and second sides of the wafer body to provide electrical shielding for the leadframe. The ground frames include ground shields providing shielding for mating ends of the corresponding signal contacts. The ground frames includes cable covers coupled to the corresponding cable shields.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
10403565	12/2018	Henry	N/A	H01R 13/6596
11081821	12/2020	Blackburn	N/A	H01R 13/6582
2014/0242841	12/2013	Trout	439/607.55	H01R 13/6585
2020/0358227	12/2019	Henry	N/A	H01R 13/6597

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
WO-2021154702	12/2020	WO	H01R 12/716

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Background/Summary

BACKGROUND OF THE INVENTION

- (1) The subject matter herein relates generally to electrical connectors for a communication system.
- (2) Communication systems use electrical connectors to electrically connect various components to allow data communication between the components. For example, electrical connectors may be directly mated together. The electrical connectors typically include a plug connector and a receptacle connector. The connectors may be cable connectors having high speed differential cables terminated to the signal conductors of the connectors. The signal conductors of the two electrical connectors transition between the two connectors. For high-speed connectors, shielding is required, adding to the complexity of the connector designs. However, as data rates increase, conventional shielding is insufficient. For example, at interfaces between the cables and the signal conductors of the connectors signal integrity problems are common.
- (3) A need remains for a reliable electrical connector having sufficient electrical shielding for high speed applications.

BRIEF DESCRIPTION OF THE INVENTION

- (4) In one embodiment, a wafer assembly for an electrical connector is provided. The wafer assembly includes a leadframe having signal contacts extend between mating ends and terminating ends. The signal contacts have main bodies between the mating ends and the terminating ends. The

wafer assembly includes cables rearward of the leadframe. Each cable includes first and second signal conductors arranged as a signal pair terminated to the terminating ends of the corresponding signal contacts. Each cable includes a cable shield providing shielding for the signal pair. The wafer assembly includes a wafer body holding the signal contacts. The wafer body has a front, a rear, a first side between the front and the rear, and a second side between the front and the rear. The cables located rearward of the rear of the wafer body. The wafer body supports the main bodies of the signal contacts. The mating ends extend forward of the front of the wafer body. The terminating ends extend rearward of the rear of the wafer body. The wafer assembly includes a first ground frame coupled to the first side of the wafer body to provide electrical shielding for the leadframe. The first ground frame includes first ground shields providing shielding for the mating ends of the corresponding signal contacts. The first ground frame includes first cable covers coupled to the corresponding cable shields. The wafer assembly includes a second ground frame coupled to the second side of the wafer body to provide electrical shielding for the leadframe. The second ground frame includes second cable covers coupled to the corresponding cable shields.

(5) In another embodiment, a wafer assembly for an electrical connector is provided. The wafer assembly includes a leadframe having signal contacts extend between mating ends and terminating ends. The signal contacts have main bodies between the mating ends and the terminating ends. The wafer assembly includes cables rearward of the leadframe. Each cable includes first and second signal conductors arranged as a signal pair terminated to the terminating ends of the corresponding signal contacts. Each cable includes a cable shield providing shielding for the signal pair. Each cable shield has a first side, a second side, a top end transitioning between the first and second sides, and a bottom end transitioning between the first and second sides. The wafer assembly includes a wafer body holding the signal contacts. The wafer body has a front, a rear, a first side between the front and the rear, and a second side between the front and the rear. The cables located rearward of the rear of the wafer body. The wafer body supports the main bodies of the signal contacts. The mating ends extend forward of the front of the wafer body. The terminating ends extend rearward of the rear of the wafer body. The wafer assembly includes a first ground frame coupled to the first side of the wafer body to provide electrical shielding for the leadframe. The first ground frame includes first ground shields providing shielding for the mating ends of the corresponding signal contacts. The first ground frame includes first cable covers coupled to the corresponding cable shields. Each first cable cover includes a first central spine, a first upper wing, and a first lower wing. The first central spine coupled to the first side of the corresponding cable shield. The first upper wing coupled to the upper end of the corresponding cable shield. The first lower wing coupled to the lower end of the corresponding cable shield. The wafer assembly includes a second ground frame coupled to the second side of the wafer body to provide electrical shielding for the leadframe. The second ground frame includes second cable covers coupled to the corresponding cable shields. Each second cable cover includes a second central spine, a second upper wing, and a second lower wing. The second central spine coupled to the second side of the corresponding cable shield. The second upper wing coupled to the upper end of the corresponding cable shield. The second lower wing coupled to the lower end of the corresponding cable shield.

(6) In a further embodiment, an electrical connector assembly is provided and includes a housing having a mating interface configured to be mated with a mating electrical connector assembly. The housing has a cavity. The electrical connector assembly includes wafer assemblies received in the cavity and coupled to the housing. The wafer assemblies are arranged in a wafer stack. Each wafer assembly includes a leadframe, a wafer body holding the leadframe, cables coupled to the leadframe, a first ground frame coupled to the wafer body to provide electrical shielding for the leadframe, and a second ground frame coupled to the wafer body to provide electrical shielding for the leadframe. The first ground frame is coupled to cable shields of each of the cables and the second ground frame is coupled to the cable shields of each of the cables. The first ground frame is

configured to be coupled to the mating electrical connector assembly and the second ground frame is configured to be coupled to the mating electrical connector assembly.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 illustrates a communication system in accordance with an exemplary embodiment.
- (2) FIG. 2 is a front perspective, partially exploded view of the first electrical connector assembly in accordance with an exemplary embodiment showing the mating interface.
- (3) FIG. 3 is a perspective view of the wafer assembly in accordance with an exemplary embodiment.
- (4) FIG. 4 is a perspective view of a portion of the wafer assembly in accordance with an exemplary embodiment, such as during an initial state of assembly.
- (5) FIG. 5 is a perspective view of a portion of the wafer assembly in accordance with an exemplary embodiment during another stage of assembly.
- (6) FIG. 6 is a perspective view of a portion of the wafer assembly in accordance with an exemplary embodiment during another stage of assembly.
- (7) FIG. 7 is a perspective view of the wafer assembly in accordance with an exemplary embodiment during another stage of assembly.
- (8) FIG. 8 is a cross-sectional view of a portion of the wafer assembly in accordance with an exemplary embodiment showing the ground frames coupled to the cable.

DETAILED DESCRIPTION OF THE INVENTION

- (9) FIG. 1 illustrates a communication system **100** in accordance with an exemplary embodiment. The communication system **100** includes a first electrical connector assembly **200** and the second electrical connector assembly **300** configured to be electrically coupled together. In various embodiments, the communication system **100** may be a server or network switch. In other various embodiments, the communication system **100** may be a backplane system. In various embodiments, the first and second electrical connector assemblies **200**, **300** are cable connector assemblies. However, in alternative embodiments, the first electrical connector assembly **200** and/or the second electrical connector assembly **300** may be a circuit board connector mounted to a circuit board.
- (10) In an exemplary embodiment, the first and second electrical connector assemblies **200**, **300** are directly mated together. For example, the first electrical connector assembly **200** may be plugged into the second electrical connector assembly **300** and/or the second electrical connector assembly **300** may be plugged into the first electrical connector assembly **200**. The first and second electrical connector assemblies **200**, **300** are mated at a separable mating interface. The first and second electrical connector assemblies **200**, **300** are directly mated together without the use of an adapter or additional electrical connector therebetween.
- (11) The first electrical connector assembly **200** includes first cables **202** terminated to a first electrical connector **204**. The first electrical connector **204** includes first signal contacts **206** and first shield structures **208** providing electrical shielding for the first signal contacts **206**.
- (12) The second electrical connector assembly **300** includes second cables **302** terminated to a second electrical connector **304**. The second electrical connector **304** includes second signal contacts **306** and second shield structures **308** providing electrical shielding for the second signal contacts **306**.
- (13) In an exemplary embodiment, the first and second electrical connectors **204**, **304** have hermaphroditic mating interfaces defined, at least in part, by the signal contacts **206**, **306** and the shield structures **208**, **308**. As such, the first and second electrical connectors **204**, **304** are identical to each other allowing use of the same parts in both the first and second electrical connectors **204**,

304. However, in alternative embodiments, the first and second electrical connectors **204**, **304** are not hermaphroditic but rather have complementary mating interfaces.

(14) In an exemplary embodiment, the signal contacts **206**, **306** are arranged in rows and columns. The first signal contacts **206** are arranged for direct mating with the second signal contacts **306** when the first and second electrical connectors **204**, **304** are mated. The shield structures **208**, **308** provide electrical shielding around the signal contacts **206**, **306** at the mating interfaces between the signal contacts **206**, **306**. In an exemplary embodiment, the first signal contacts **206** and the first shield structures **208** are pluggable into the second electrical connector **304**. The second signal contacts **306** and the second shield structures **308** are pluggable into the first electrical connector **204**. The communication system **100** is a direct plug communication system.

(15) The signal contacts **206**, **306** define electrical paths between the cables **202**, **302**. The signal contacts **206**, **306** mate at a separable mating interface between the first and second electrical connectors **204**, **304**. For example, the mating interfaces of the signal contacts **206**, **306** are arranged along mating planes (for example, parallel to the columns). In various embodiments, the first signal contacts **206** are arranged in pairs and the second signal contacts **306** are arranged in pairs. The shield structures **208**, **308** cooperate to provide shielding for the corresponding signal contacts **206**, **306** (for example, pairs of the signal contacts **206**, **306**). In an exemplary embodiment, the shield structures **208**, **308** provide 360° shielding for the corresponding signal contacts **206**, **306**. The shield structures **208**, **308** may be electrically connected to cable shields of the cables **202**, **302** to continue shielding along the signal paths between the cables **202**, **302**. The shield structures **208**, **308** may be electrically connected to shielding structures passing through the electrical connectors **204**, **304**.

(16) The first electrical connector **204** includes a housing **210** having a mating interface configured to be mated with the second electrical connector **304**. The mating interface is provided at a front of the housing **210**. In an exemplary embodiment, the first electrical connector **204** includes a plurality of wafer assemblies **230** coupled to the housing **210**. The wafer assemblies **230** are received in a cavity **209** of the housing **210**. The wafer assemblies **230** include the signal contacts **206** and the shield structures **208**. The cables **202** are configured to be terminated to corresponding wafer assemblies **230**. For example, the wafer assemblies **230** may support the cables **202** and signal conductors of the cables **202** are soldered or otherwise terminated to corresponding signal contacts **206**. The cables **202** may extend into the cavity **209**. In an exemplary embodiment, the wafer assemblies **230** are oriented vertically. However, other orientations are possible in alternative embodiments. Each wafer assembly **230** includes a corresponding column of the signal contacts **206**. The wafer assemblies **230** are stacked in the housing **210** to arrange the signal contacts **206** in rows.

(17) In an exemplary embodiment, the wafer assemblies **230** are arranged in a wafer stack **232**. For example, the wafer assemblies **230** are parallel to each other in the wafer stack **232**. The wafer stack **232** extends from a rear of the housing **210**. Optionally, the wafer assemblies **230** may be individually loaded into the housing **210**, such as into the cavity **209** at a rear of the housing **210**. Alternatively, the wafer assemblies **230** may be assembled together in the wafer stack **232**, and the wafer stack **232** is loaded into the rear of the housing **210**.

(18) In an exemplary embodiment, each wafer assembly **230** extends between a mating end **234** and a terminating end **236**. The cables **202** are terminated to the wafer assembly **230** at the terminating end **236**. The mating end **234** extends into the housing **210** and is configured to be mated with the second electrical connector **304**. In various embodiments, the wafer assembly **230** may be a right-angle wafer assembly having the mating end **234** at a right angle relative to the terminating end **236**. The shield structures **208** are provided at the mating end **234** and are configured to be mated with the second electrical connector **304**.

(19) The second electrical connector **304** includes a housing **310** having a mating interface configured to be mated with the first electrical connector **204**. The mating interface is provided at a

front of the housing **310**. In an exemplary embodiment, the second electrical connector **304** includes a plurality of wafer assemblies **330** coupled to the housing **310**. The wafer assemblies **330** are received in a cavity **309** of the housing **210**. The wafer assemblies **330** include the signal contacts **306** and the shield structures **308**. The cables **302** are terminated to the corresponding wafer assemblies **330**. For example, conductors of the cables **302** may be soldered or welded to the signal contacts **306**. The cables **302** may extend into the cavity **309**. In an exemplary embodiment, the wafer assemblies **330** are vertically. However, other orientations are possible in alternative embodiments. Each wafer assembly **330** includes a corresponding column of the signal contacts **306**. The wafer assemblies **330** are stacked in the housing **310** to arrange the signal contacts **306** in rows.

(20) In an exemplary embodiment, the wafer assemblies **330** are arranged in a wafer stack **332**. For example, the wafer assemblies **330** are parallel to each other in the wafer stack **332**. The wafer stack **332** extends from a rear of the housing **310**. Optionally, the wafer assemblies **330** may be individually loaded into the housing **310**, such as into the cavity **309** at a rear of the housing **310**. Alternatively, the wafer assemblies **330** may be assembled together in the wafer stack **332** and the wafer stack **332** is loaded into the rear of the housing **310**.

(21) In an exemplary embodiment, each wafer assembly **330** extends between a mating end **334** and a terminating end **336**. The cables **302** are terminated to the wafer assembly **330** at the terminating end **336**. The mating end **334** extends into the housing **310** is configured to be mated with the first electrical connector **204**. In various embodiments, the wafer assembly **330** may be a right-angle wafer assembly having the mating end **334** at a right angle relative to the terminating end **336**. The shield structures **308** are provided at the mating end **334** and are configured to be mated with the first electrical connector **204**.

(22) FIG. 2 is a front perspective, partially exploded view of the first electrical connector assembly **200** in accordance with an exemplary embodiment showing the mating interface. The second electrical connector assembly **300** (FIG. 1) may have an identical mating interface. One of the wafer assemblies **230** is shown poised for loading into the housing **210**. The housing **210** holds the signal contacts **206** and the shield structures **208** for mating with the second electrical connector **304** (shown in FIG. 1). The housing **210** forms part of the mating interface with the second electrical connector **304**.

(23) The housing **210** has a top **211** and a bottom **212**. The housing **210** is a first side **213** and a second side **214** opposite the first side **213**. The housing **210** has a primary axis **215** extending from top **211** to bottom **212** and a secondary axis **216** extending from the first side **213** to the second side **214**. The secondary axis **216** is perpendicular to the primary axis **215**. In an exemplary embodiment, the signal contacts **206** and the shield structures **208** are arranged in columns parallel to the primary axis **215** and rows parallel to the secondary axis **216**. The mating ends **234** are arranged along mating planes parallel to the primary axis **215** for interfacing with the second contacts **306** (FIG. 1). The wafer assemblies **230** are received in the housing **210** such that the wafer assemblies **230** are oriented parallel to the primary axis **215**.

(24) In an exemplary embodiment, the housing **210** is a multi-piece housing including a contact organizer **217** and a commoning member **218**. The commoning member **218** is at the front of the housing **210**. The contact organizer **217** may include locating features for locating the commoning member **218** relative to the contact organizer **217**. In an exemplary embodiment, the commoning member **218** faces the second electrical connector **304**. The commoning member **218** is electrically conductive and is used to electrically common each of the shield structures **208**. The commoning member **218** provides electrical shielding for the signal contacts **206** at the mating interface. The commoning member **218** may be electrically connected to the shield structures **308** (shown in FIG. 1) of the second electrical connector **304**.

(25) In an exemplary embodiment, the contact organizer **217** includes a base **219**, an outer shroud **221** surrounding the cavity **209**. The base **219** may support towers **220** in the cavity **209**. The

towers **220** extend forward from the base **219**. The towers **220** may be integral with the base **219**, such as being co-molded with the base **219**. In alternative embodiments, the towers **220** may be separate from the base **219** and loaded into the base **219**. For example, the towers **220** may be part of the wafer assemblies **230**. The towers **220** support the signal contacts **206** and the shield structures **208**. In an exemplary embodiment, the towers **220** extend into openings **222** in the commoning member **218**. The towers **220** may pass entirely through the openings **222** and extend forward of the front of the commoning member **218**. The towers **220** are configured to be received in corresponding openings in a commoning member of the second electrical connector **304**.

(26) The wafer assemblies **230** are coupled to the housing **210** rearward of the base **219**. The signal contacts **206** and the shield structures **208** pass through the base **219** to extend along the towers **220**. The signal contacts **206** are electrically isolated from each other and from the shield structures **208** by the dielectric material of the towers **220**.

(27) The commoning member **218** is manufactured from a conductive material. For example, the commoning member **218** may be a metal block having the openings **222** formed therethrough. In alternative embodiments, the commoning member **218** may be manufactured from a conductive plastic. In other various embodiments, the commoning member **218** may be a plated plastic structure having plating at the front **224** and/or through the openings **222** and/or at the rear. The shield structures **208** are configured to be electrically connected to the commoning member **218**. For example, the shield structures **208** may engage the commoning member **218** within the openings **222**.

(28) In an exemplary embodiment, the openings **222** pass entirely through the commoning member **218** and are defined by walls **225**. In an exemplary embodiment, the openings **222** are rectangular. In the illustrated embodiment, the openings **222** are square shaped. However, the openings **222** may have other shapes. In an exemplary embodiment, the openings **222** are oversized relative to the towers **220**. For example, each opening **222** may be sized to receive two of the towers **220** (one from the first electrical connector **204** and one from the second electrical connector **304**).

(29) FIG. **3** is a perspective view of the wafer assembly **230** in accordance with an exemplary embodiment. In an exemplary embodiment, the wafer assembly **230** is identical to the wafer assembly **330** (shown in FIG. **1**) with both wafer assemblies **230**, **330** including identical components.

(30) The wafer assembly **230** includes the cables **202**, the signal contacts **206** and the shield structures **208**. The signal contacts **206** and the shield structures **208** are terminated to the cables **202**. In an exemplary embodiment, the wafer assembly **230** includes a leadframe **240** including the signal contacts **206**. The wafer assembly **230** includes a dielectric wafer body **242** holding the leadframe **240**. The wafer assembly **230** includes a first ground frame **600** coupled to a first side of the wafer body **242** and a second ground frame **602** coupled to a second side of the wafer body **242**. The first and second ground frames **600**, **602** form the shield structures **208**. The first and second ground frames **600**, **602** provide electrical shielding for the leadframe **240**. The first and second ground frames **600**, **602** are electrically connected to the cables **202**.

(31) In an exemplary embodiment, the wafer assembly **230** includes a wafer frame **231** having a cavity **233**. The wafer body **242** and the cables **202** are received in the cavity **233**. The wafer frame **231** is used to secure the wafer assembly **230** in the housing **210** (shown in FIG. **2**). The wafer frame **231** includes latches **235** to latchably secure the wafer frame **231** in the housing **210**.

(32) In an exemplary embodiment, the wafer assembly **230** includes a cable holder **203** holding the cables **202**. The cable holder **203** is received in the cavity **233**. In various embodiments, the cable holder **203** is an overmolded body that is overmolded over the cables **202**. The cable holder **203** may be formed in place on the cables **202**. The cable holder **203** may be formed in place in the cavity **233**.

(33) FIG. **4** is a perspective view of a portion of the wafer assembly **230** in accordance with an exemplary embodiment, such as during an initial state of assembly. FIG. **4** shows the cables **202**,

the leadframe **240**, the dielectric wafer body **242**, and the towers **220**. The cables **202** are shown terminated to the signal contacts **206** of the leadframe **240**. The wafer body **242** supports the signal contacts **206** of the leadframe **240**. The towers **220** extend from the wafer body **242** to support ends of the signal contacts **206**.

(34) The cables **202** are shielded cables. In an exemplary embodiment, each cable **202** is a twin-axial cable having a pair of signal conductors, namely a first signal conductor **500** and a second signal conductor **502**. The signal conductors **500**, **502** are arranged as a signal pair. The signal conductors **500**, **502** are held by an insulator **504**. A cable shield **506** surrounds the insulator **504**. The cable shield **506** provides shielding for the signal pair of signal conductors **500**, **502** along the length of the cable **202**. A cable jacket **508** surrounds the cable shield **506**. The end of the cable **202** is stripped for termination to the signal contacts **206**.

(35) In an exemplary embodiment, the cable **202** is oval shaped to surround the signal pair of signal conductors **500**, **502**. In various embodiments, the cable shield **506** includes a first side **510** and a second side **512** opposite the first side **510**. The cable shield **506** includes a first or upper end **514** and a second or lower end **516** opposite the first end **514**. Optionally, the sides **510**, **512** may be generally flat or planar. The ends **514**, **516** are curved between the sides **510**, **512**. The cable **202** may have other shapes in alternative embodiments, such as having the sides **510**, **512** curved, such as at a radius of curvature that is different from the radius of curvature of the ends **514**, **516**.

(36) The leadframe **240** is a stamped and formed leadframe that forms the signal contacts **206** from a metal sheet. In an exemplary embodiment, the leadframe **240** only includes the signal contacts **206**. However, in alternative embodiments, the leadframe **240** may include ground contacts arranged between corresponding signal contacts to provide electrical shielding for the signal contacts. In an exemplary embodiment, the signal contacts **206** are arranged in pairs configured to carry differential signals. However, the signal contacts **206** may be single ended signal contacts in alternative embodiments.

(37) The wafer body **242** surrounds the signal contacts **206** and positions the signal contacts **206** relative to each other. In an exemplary embodiment, the wafer body **242** is manufactured from a dielectric material, such as a plastic material. In an exemplary embodiment, the wafer body **242** is an overmold that is overmolded around the leadframe **240**. The wafer body **242** includes first and second sides **250**, **252**. The wafer body **242** includes a front **254** and a rear **256** extending between a top and a bottom. The front **254** defines a mating end. The signal contacts **206** extend from the wafer body **242** at the front **254** for connection to the second electrical connector **304** (shown in FIG. 1). In an exemplary embodiment, the towers **220** extend from the front **254** of the wafer body **242** to support the ends of the signal contacts **206**. In the illustrated embodiment, the towers **220** are separate and discrete from the wafer body **242** and coupled thereto. In alternative embodiments, the towers **220** are integral with the wafer body **242**, such as being co-molded with the wafer body **242**. The rear **256** defines a cable end. The cables **202** extend from the rear **256** along cable axes.

(38) Each signal contact **206** includes a contact body **270** extending between a mating end **272** and a terminating end **274**. The contact body **270** extends along a contact axis. Optionally, the contact axis is parallel to the corresponding cable axis. In an exemplary embodiment, the contact body **270** is stamped and formed as part of the leadframe **240**. The contact bodies **270** of the leadframe **240** are generally arranged in a leadframe plane parallel to the sides **250**, **252** of the wafer body **242**. The signal contact **206** includes a spring beam **276** at the mating end **272**. The spring beam **276** is deflectable and configured to be mated with a corresponding spring beam of the second signal contact **306** (shown in FIG. 1). In various embodiments, the spring beam **276** is generally aligned with the main portion of the contact body **270**. In an exemplary embodiment, the signal contact **206** includes a solder pad **278** at the terminating end **274** for soldering or welding to the corresponding signal conductor **500**, **502** of the cable **202**.

(39) FIG. 5 is a perspective view of a portion of the wafer assembly **230** in accordance with an exemplary embodiment during another stage of assembly. FIG. 5 shows the cables **202**, the

leadframe **240**, the dielectric wafer body **242**, the towers **220**, conductor supports **280**, and the second ground frame **602**. After the conductors **500**, **502** of the cables **202** terminated to the solder pads **278** of the signal contacts **206**, the conductor supports **280** are added to support the termination. The conductor supports **280** provide strain relief for the conductors **500**, **502**. In various embodiments, the conductor supports **280** are molded in place over the conductors **500**, **502** and the solder pads **278**.

(40) In an exemplary embodiment, the second ground frame **602** is coupled to the second side **252** of the wafer body **242**. The second ground frame **602** extends along the towers **220** to provide shielding for the spring beams **276** of the signal contacts **206**. The second ground frame **602** is configured to be electrically connected to the second side **512** of the cable shield **506** to provide shielding along the signal paths to the cables **202**. In an exemplary embodiment, the second ground frame **602** is configured to be electrically connected to the cable shields **506** of each of the cables **202**.

(41) FIG. **6** is a perspective view of a portion of the wafer assembly **230** in accordance with an exemplary embodiment during another stage of assembly. FIG. **6** shows the first ground frame **600** coupled to the wafer body **242** and the cables **202**. The first ground frame **600** is coupled to the first side **250** of the wafer body **242**. The first ground frame **600** extends along the towers **220** to provide shielding for the spring beams **276** of the signal contacts **206**. The first ground frame **600** is configured to be electrically connected to the first side **510** of the cable shield **506** to provide shielding along the signal paths to the cables **202**. In an exemplary embodiment, the first ground frame **600** is configured to be electrically connected to the cable shields **506** of each of the cables **202**.

(42) The first ground frame **600** may be similar to the second ground frame **602** and both may include like components identified with like reference numerals. The ground frame **600** provides a shield structure for the signal contacts **206**. In an exemplary embodiment, the ground frame **600** includes a ground plate **610**, ground shields **620** extending from the ground plate **610** for shielding the mating ends of the signal contacts **206**, and cable covers **640** extending from the ground plate **610** for connection to the cable shields **506**.

(43) The ground plate **610** forms a main body of the ground frame **600**. The ground plate **610** may be planar. The ground plate **610** is coupled to the wafer body **242**, such as to the first side **250** of the wafer body **242**. The ground plate **610** may include openings **612** that receive bosses **258** extending from the wafer body **242**. The bosses **258** may be heat stacked to secure the ground plate **610** to the wafer body **242**. The ground plate **610** extends between a front edge **614** and a rear edge **616**. The ground shields **620** extend from the front edge **614** of the ground plate **610** and the cable covers **640** extend from the rear edge **616** of the ground plate **610**. The ground plate **610** extends between a top and a bottom.

(44) Each ground shield **620** includes a shield portion **630** and a transition portion **632** between the shield portion **630** and the ground plate **610**. The shield portion **630** provides electrical shielding along the mating ends **272** of the signal contacts **206**. The transition portion **632** may include one or more bends to position the shield portion **630** relative to the ground plate **610**. In an exemplary embodiment, the ground frame **600** includes connecting beams **622** extending forward from the front edge **614**. The connecting beams **622** are configured to be electrically connected to the commoning member **218** (shown in FIG. **2**). The connecting beams **622** may be deflectable spring beams.

(45) In the illustrated embodiment, the shield portion **630** of the ground shield **620** is C-shaped. The shield portion **630** includes an end wall **634** and side walls **636**, **638** extending from the end wall **634**. The transition portion **632** is connected to the end wall **634**. In an exemplary embodiment, the shield portion **630** is stamped such that the end wall **634** includes one or more ground fingers **635** and such that the side walls **636**, **638** include one or more ground fingers **637**, **639**, respectively. The ground fingers **635**, **637**, **639** include mating interfaces. For example, the

ground fingers **635, 637, 639** may be cupped or include bumps near distal ends of the ground fingers **635, 637, 639**. The ground fingers **635, 637, 639** are deflectable. Optionally, the end wall **634** and/or the side walls **636, 638** may include dimples **633**. The ground fingers **635, 637, 639** extend along the mating ends **272** of the signal contacts **206**.

(46) When assembled, as shown in FIG. **6**, the ground plate(s) **610** extend along the wafer body **242**. The ground shields **620** extend forward of the wafer body **242** to extend along the mating ends **272** of the signal contacts **206**. The shield portions **630** of the ground shields **620** provide shielding for the corresponding pair of signal contacts **206**. The ground shields **620** are C-shaped and surround three sides of the pair of signal contacts **206**. The ground fingers **635** have generally uniform spacing from the spring beams **276** of the pair. The ground fingers **637, 639** of the side walls **636, 638** are spaced generally uniformly from the respective (closest) signal contact **206**. The ground shields **620** provide efficient electrical shielding for both signal contacts **206** of the corresponding pair.

(47) Each cable cover **640** extends rearward from the ground plate **610** to couple to the cable shield **506** of the corresponding cable **202**. The cable cover **640** may be soldered to the cable shield **506** to create a direct electrical connection. In an exemplary embodiment, the cable cover **640** includes a solder pad **642** configured to be soldered to the cable shield **506**. The solder pad **642** includes a solder pocket **644** that receives solder to couple the cable cover **640** to the cable shield **506**. The solder pocket **644** may be an opening through the cable cover **640**. In an exemplary embodiment, the cable cover **640** includes one or more thermal relief openings **646** adjacent the solder pocket **644**. The thermal relief openings **646** limit the heating needed for soldering the solder pad **642** to the cable shield **506**. The thermal relief openings **646** allow the heat to be focused at the solder pocket **644** and the solder during the soldering process reducing the soldering time and energy needed to perform the soldering process. Optionally, the cable cover **640** may be soldered at multiple points along the cable shield **506**.

(48) In an exemplary embodiment, the cable cover **640** includes a central spine **650**, a first upper wing **652** at an upper side of the central spine **650**, and a lower wing **654** at a lower side of the central spine **650**. The central spine **650** extends to a distal end **656** at a rear of the cable cover **640**. The central spine **650** defines the solder pad **642**. The central spine **650** extends along the first side **510** of the cable shield **506**. The central spine **650** is coupled to the first side **510** of the corresponding cable shield **506**. For example, the central spine **650** is soldered to the first side **510** for a direct electrical connection. Portions of the central spine **650** may be capacitively coupled to the first side **510**, such as due to close proximity of the central spine **650** to the first side **510**. The upper wing **652** is coupled to the upper end **514** of the corresponding cable shield **506**. For example, the upper wing **652** may be capacitively coupled to the upper end **514**, such as due to close proximity of the upper wing **652** to the upper end **514**. In an exemplary embodiment, the upper wing **652** has a curved profile to follow a curvature of the cable shield **506** to stay in close proximity to the cable shield **506**. The lower wing **654** is coupled to the lower end **516** of the corresponding cable shield **506**. For example, the lower wing **654** may be capacitively coupled to the lower end **516**, such as due to close proximity of the lower wing **654** to the lower end **516**. In an exemplary embodiment, the lower wing **654** has a curved profile to follow a curvature of the cable shield **506** to stay in close proximity to the cable shield **506**.

(49) FIG. **7** is a perspective view of the wafer assembly **230** in accordance with an exemplary embodiment during another stage of assembly. FIG. **7** shows the cables, the ground frames **600, 602**, the wafer body **242**, and the leadframe **240** in the cavity **233** of the wafer frame **231**.

(50) In an exemplary embodiment, the wafer frame **231** includes separating walls **237** between the cables **202**. The cavity **233** of the wafer frame **231** is configured to receive the cable holder **203** (shown in FIG. **3**). The wafer frame **231** and the cable holder **203** are used to hold the cables **202** relative to each other and may provide strain relief for the cables **202**. The ground frames **600, 602** are both electrically connected to the cables **202**, such as to opposite sides of the cables **202**, to

provide multiple ground paths between the shield structure **208** and the cable shields **506** of the cables **202**. The ground frames **600**, **602** extend along the leadframe **240** to provide shielding for the signal contacts **206**. The ground shields **620** of the ground frames **600**, **602** provide shielding along the mating ends **272** of the signal contacts **206**. The ground shields **620** are configured for mating with the second electrical connector assembly **300** (shown in FIG. 1).

(51) FIG. 8 is a cross-sectional view of a portion of the wafer assembly **230** in accordance with an exemplary embodiment showing the ground frames **600**, **602** coupled to the cable **202**. The first ground frame **600** is coupled to the first side **250** of the wafer body **242** and the second ground frame **602** is coupled to the second side **252** of the wafer body **242**. The first cable cover **640a** extends along the first side **510** of the cable shield **506** and the second cable cover **640b** extends along the second side **512** of the cable shield **506**.

(52) The first and second cable covers **640a**, **640b** cooperate to form a cable shield pocket **660**. The cable **202** is located in the cable shield pocket **660**. In an exemplary embodiment, the first and second cable covers **640a**, **640b** surround a majority of the cable shield **506** to provide circumferential shielding and connection at the interface between the ground frames **600**, **602** and the cable **202**. For example, the first and second cable covers **640a**, **640b** may surround (for example, cover or extend along) at least 50% of the cable shield **506**. In various embodiments, the first and second cable covers **640a**, **640b** may surround (for example, cover or extend along) at least 75% of the cable shield **506**. The gap or spacing between the ends of the first and second cable covers **640a**, **640b** is relatively small (for example, narrow width) when compared to the distance from one neighboring signal pair to the next. Having the cable covers **640a**, **640b** close together relative to the distance to the next signal pair reduces cross-talk, even without having the material of the cable covers **640a**, **640b** across the gap or space. In an exemplary embodiment, the cable covers **640a**, **640b** may have multiple points of electrical connection with the cable **202**. The cable covers **640a**, **640b** may be capacitively coupled to the cable **202** in areas that are not directly electrically connected to the cable **202**.

(53) In an exemplary embodiment, the central spine **650a** of the first ground frame **600** is coupled to the first side **510** of the corresponding cable shield **506**, such as being soldered to the first side **510** for a direct electrical connection. Portions of the central spine **650a** may be capacitively coupled to the first side **510**, such as due to close proximity of the central spine **650a** to the first side **510**. The upper wing **652a** of the first ground frame **600** is coupled to the upper end **514** of the corresponding cable shield **506**, such as being capacitively coupled to the upper end **514** due to the close proximity of the upper wing **652a** to the upper end **514**. The curved profile of the upper wing **652a** closely follows the curvature of the cable shield **506** to stay in close proximity to the cable shield **506**. The lower wing **654a** of the first ground frame **600** is coupled to the lower end **516** of the corresponding cable shield **506**, such as being capacitively coupled to the lower end **516** due to the close proximity of the lower wing **654a** to the lower end **516**. The curved profile of the lower wing **654a** closely follows the curvature of the cable shield **506** to stay in close proximity to the cable shield **506**. In an exemplary embodiment, the central spine **650a** is located outside of an outer surface **617** of the ground plate **610**. The upper and lower wings **652a**, **654a** are curved relative to the central spine **650a**. For example, the upper and lower wings **652a**, **654a** are curved to extend inside of an inner surface **618** of the ground plate **610**.

(54) In an exemplary embodiment, the central spine **650b** of the second ground frame **602** is coupled to the second side **512** of the corresponding cable shield **506**, such as being soldered to the second side **512** for a direct electrical connection. Portions of the central spine **650b** may be capacitively coupled to the second side **512**, such as due to close proximity of the central spine **650b** to the second side **512**. The upper wing **652b** of the second ground frame **602** is coupled to the upper end **514** of the corresponding cable shield **506**, such as being capacitively coupled to the upper end **514** due to the close proximity of the upper wing **652b** to the upper end **514**. The curved profile of the upper wing **652b** closely follows the curvature of the cable shield **506** to stay in close

proximity to the cable shield **506**. The lower wing **654b** of the second ground frame **602** is coupled to the lower end **516** of the corresponding cable shield **506**, such as being capacitively coupled to the lower end **516** due to the close proximity of the lower wing **654b** to the lower end **516**. The curved profile of the lower wing **654b** closely follows the curvature of the cable shield **506** to stay in close proximity to the cable shield **506**.

(55) It is to be understood that the above description is intended to be illustrative, and not restrictive. For example, the above-described embodiments (and/or aspects thereof) may be used in combination with each other. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the invention without departing from its scope.

Dimensions, types of materials, orientations of the various components, and the number and positions of the various components described herein are intended to define parameters of certain embodiments, and are by no means limiting and are merely exemplary embodiments. Many other embodiments and modifications within the spirit and scope of the claims will be apparent to those of skill in the art upon reviewing the above description. The scope of the invention should, therefore, be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled. In the appended claims, the terms “including” and “in which” are used as the plain-English equivalents of the respective terms “comprising” and “wherein.” Moreover, in the following claims, the terms “first,” “second,” and “third,” etc. are used merely as labels, and are not intended to impose numerical requirements on their objects. Further, the limitations of the following claims are not written in means-plus-function format and are not intended to be interpreted based on 35 U.S.C. § 112(f), unless and until such claim limitations expressly use the phrase “means for” followed by a statement of function void of further structure.

Claims

1. A wafer assembly for an electrical connector, the wafer assembly comprising: a leadframe having signal contacts extending between mating ends and terminating ends, the signal contacts having main bodies between the mating ends and the terminating ends; cables rearward of the leadframe, each cable including first and second signal conductors arranged as a signal pair, the first and second signal conductors being terminated to the terminating ends of the corresponding signal contacts at termination regions, each cable includes a cable shield providing shielding for the signal pair; a wafer body holding the signal contacts, the wafer body having a front, a rear, a first side between the front and the rear, and a second side between the front and the rear, the cables and termination regions located rearward of the rear of the wafer body, wherein the wafer body supports the main bodies of the signal contacts, the mating ends extending forward of the front of the wafer body, the terminating ends extending rearward of the rear of the wafer body into the termination regions; a first ground frame coupled to the first side of the wafer body to provide electrical shielding for the leadframe, the first ground frame including first ground shields providing shielding for the mating ends of the corresponding signal contacts, the first ground frame including first cable covers covering the termination regions, the first cable covers coupled to the corresponding cable shields; and a second ground frame coupled to the second side of the wafer body to provide electrical shielding for the leadframe, the second ground frame including second cable covers covering the termination regions, the second cable covers coupled to the corresponding cable shields.

2. The wafer assembly of claim 1, wherein the first cable covers are coupled to the cable shields of all of the cables to electrically connect each of the cable shields, and wherein the second cable covers are coupled to the cable shields of all of the cables to electrically connect each of the cable shields.

3. The wafer assembly of claim 1, wherein the first cable cover has a curved profile following a curvature of the corresponding cable shield, and wherein the second cable cover has a curved

profile following a curvature of the corresponding cable shield.

4. The wafer assembly of claim 1, wherein each first cable cover includes a solder pad soldered to the corresponding cable shield, portions of the first cable cover surrounding the solder pad being positioned in close proximity to the corresponding cable shield and being capacitively coupled to the corresponding cable shield, and wherein each second cable cover includes a solder pad soldered to the corresponding cable shield, portions of the second cable cover surrounding the solder pad being positioned in close proximity to the corresponding cable shield and being capacitively coupled to the corresponding cable shield.

5. The wafer assembly of claim 1, wherein each first cable cover includes a first central spine, a first upper wing, and a first lower wing, the first central spine being coupled to a first side of the corresponding cable shield, the first upper wing being coupled to an upper end of the corresponding cable shield, the first lower wing being coupled to a lower end of the corresponding cable shield, and wherein each second cable cover includes a second central spine, a second upper wing, and a second lower wing, the second central spine being coupled to a second side of the corresponding cable shield, the second upper wing being coupled to an upper end of the corresponding cable shield, the second lower wing being coupled to a lower end of the corresponding cable shield.

6. The wafer assembly of claim 5, wherein the first upper wing is capacitively coupled to the upper end of the corresponding cable shield and the first lower wing is capacitively coupled to the lower end of the corresponding cable shield, and wherein the second upper wing is capacitively coupled to the upper end of the corresponding cable shield and the second lower wing is capacitively coupled to the lower end of the corresponding cable shield.

7. The wafer assembly of claim 5, wherein the first central spine is planar, the first upper wing being curved to follow a curvature of the corresponding cable shield, the first lower wing being curved to follow a curvature of the corresponding cable shield, and wherein the second central spine is planar, the second upper wing being curved to follow a curvature of the corresponding cable shield, the second lower wing being curved to follow a curvature of the corresponding cable shield.

8. The wafer assembly of claim 5, wherein the first ground frame includes a first ground plate, the first ground shields extending forward of the first ground plate, the first cable covers extending rearward of the first ground frame, and wherein the second ground frame includes a second ground plate, the second ground shields extending forward of the second ground plate, the second cable covers extending rearward of the second ground frame.

9. The wafer assembly of claim 8, wherein the first ground plate includes an inner surface facing the wafer body and an outer surface opposite the inner surface, the first cable cover located outside of the outer surface, the first upper wing and the first lower wing being located inside of the inner surface, and wherein the second ground plate includes an inner surface facing the wafer body and an outer surface opposite the inner surface, the second cable cover located outside of the outer surface, the second upper wing and the second lower wing being located inside of the inner surface.

10. The wafer assembly of claim 1, wherein each first cable cover includes a solder pocket receiving solder to couple the first cable cover to the cable shield, the first cable cover including a thermal relief opening adjacent the solder pocket, and wherein each second cable cover includes a solder pocket receiving solder to couple the second cable cover to the cable shield, the second cable cover including a thermal relief opening adjacent the solder pocket.

11. The wafer assembly of claim 1, wherein the first and second cable covers cooperate to form a cable shield pocket surrounding a majority of the corresponding cable shield.

12. The wafer assembly of claim 11, wherein the first and second cable covers surround greater than 75% of the corresponding cable shield.

13. The wafer assembly of claim 1, wherein the second ground frame includes second cable covers providing shielding for the mating ends of the corresponding signal contacts.

14. A wafer assembly for an electrical connector, the wafer assembly comprising: a leadframe

having signal contacts extending between mating ends and terminating ends, the signal contacts having main bodies between the mating ends and the terminating ends; cables rearward of the leadframe, each cable including first and second signal conductors arranged as a signal pair, the first and second signal conductors terminated to the terminating ends of the corresponding signal contacts, each cable includes a cable shield providing shielding for the signal pair, each cable shield having a first side, a second side, a top end transitioning between the first and second sides, and a bottom end transitioning between the first and second sides; a wafer body holding the signal contacts, the wafer body having a front, a rear, a first side between the front and the rear, and a second side between the front and the rear, the cables located rearward of the rear of the wafer body, wherein the wafer body supports the main bodies of the signal contacts, the mating ends extending forward of the front of the wafer body, the terminating ends extending rearward of the rear of the wafer body; a first ground frame coupled to the first side of the wafer body to provide electrical shielding for the leadframe, the first ground frame includes a first ground plate along the first side of the wafer body, the first ground frame including first ground shields extending forward of the first ground plate and providing shielding for the mating ends of the corresponding signal contacts, the first ground frame including first cable covers extending rearward of the first ground plate and coupled to the corresponding cable shields, each first cable cover including a first central spine, a first upper wing, and a first lower wing, the first central spine coupled to the first side of the corresponding cable shield, the first upper wing coupled to the upper end of the corresponding cable shield, the first lower wing coupled to the lower end of the corresponding cable shield; and a second ground frame coupled to the second side of the wafer body to provide electrical shielding for the leadframe, the second ground frame includes a second ground plate along the second side of the wafer body, the second ground frame including second cable covers extending rearward of the second ground plate and coupled to the corresponding cable shields, each second cable cover including a second central spine, a second upper wing, and a second lower wing, the second central spine coupled to the second side of the corresponding cable shield, the second upper wing coupled to the upper end of the corresponding cable shield, the second lower wing coupled to the lower end of the corresponding cable shield.

15. The wafer assembly of claim 14, wherein the first cable covers are coupled to the cable shields of all of the cables to electrically connect each of the cable shields, and wherein the second cable covers are coupled to the cable shields of all of the cables to electrically connect each of the cable shields.

16. The wafer assembly of claim 14, wherein each first cable cover has a curved profile following a curvature of the corresponding cable shield, and wherein each second cable cover has a curved profile following a curvature of the corresponding cable shield.

17. The wafer assembly of claim 14, wherein each first central spine includes a solder pad soldered to the corresponding cable shield, each first upper wing and the first lower wing positioned in close proximity to the corresponding cable shield and being capacitively coupled to the corresponding cable shield, and wherein each second central spine includes a solder pad soldered to the corresponding cable shield, each second upper wing and the second lower wing positioned in close proximity to the corresponding cable shield and being capacitively coupled to the corresponding cable shield.

18. The wafer assembly of claim 14, wherein the first upper wing is capacitively coupled to the upper end of the corresponding cable shield and the first lower wing is capacitively coupled to the lower end of the corresponding cable shield, and wherein the second upper wing is capacitively coupled to the upper end of the corresponding cable shield and the second lower wing is capacitively coupled to the lower end of the corresponding cable shield.

19. The wafer assembly of claim 14, wherein the first central spine is planar, the first upper wing being curved to follow a curvature of the corresponding cable shield, the first lower wing being curved to follow a curvature of the corresponding cable shield, and wherein the second central

spine is planar, the second upper wing being curved to follow a curvature of the corresponding cable shield, the second lower wing being curved to follow a curvature of the corresponding cable shield.

20. An electrical connector assembly comprising: a housing having a mating interface configured to be mated with a mating electrical connector assembly, the housing having a cavity; and wafer assemblies received in the cavity and coupled to the housing, the wafer assemblies arranged in a wafer stack, each wafer assembly including a leadframe, a wafer body holding the leadframe, cables coupled to the leadframe at termination regions, a first ground frame coupled to the wafer body to provide electrical shielding for the leadframe, and a second ground frame coupled to the wafer body to provide electrical shielding for the leadframe, wherein the first ground frame covers the termination regions and is coupled to cable shields of each of the cables and the second ground frame covers the termination regions and is coupled to the cable shields of each of the cables, and wherein the first ground frame is configured to be coupled to the mating electrical connector assembly and the second ground frame is configured to be coupled to the mating electrical connector assembly.
